Aymen Danoun, Ph.D.

Mechanical & Aerospace Engineer | AI & ML | FEA/CAE | Structural Analysis | Computational Mechanics
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Professional Summary

PhD-qualified Mechanical and Aerospace Engineer with 3 years of industrial aerospace experience delivering structural engineering work for Airbus aircraft programs. Specialized in Finite Element Analysis (FEA), Computational Mechanics, Aerospace and composite structures, and nonlinear multiscale simulations. Experienced in developing hybrid AI/ML and physics-based models for complex engineering problems. Strong background in CAE, structural optimization, multiscale modeling, and simulation acceleration using deep learning.

Professional Experience

Aerospace Engineer **2023–2026**
Accenture — Aerospace Engineering Projects for Airbus Toulouse, France

- Performed **FEA and structural analysis of multiple aerospace fuselage sections** using HyperMesh and MSC Nastran to support aircraft engineering and R&D activities.
- Developed a new structural door-frame concept, known as the **Torquebox concept**, for multiple aircraft sections supporting future **Airbus A320** programs; managed **composite structural models** including GFEM integration, model condensation, stress analysis, and post-processing.
- Achieved an overall **10% mass reduction** through **design and mass optimization** of the Torquebox door-frame for the **A320 nose fuselage and section S14A**, accounting for multiple composite-material sizing criteria.
- Built a **parametric FEA workflow in Altair HyperWorks** to automate CAE model generation and support repeatable aerospace analysis.
- Modeled **anodization processes** and investigated their application to improve aeronautical structural performance.
- Contributed to Airbus engineering activities in stress analysis and structural design working with multidisciplinary teams.

Doctoral Researcher **2019–2022**
Institut de Mécanique et d'Ingénierie de Bordeaux (I2M) Bordeaux, France

- Developed **hybrid Artificial Intelligence and physics-based modeling methods** for nonlinear numerical simulations of multiscale heterogeneous and architected materials.
- Integrated **deep learning models into Finite Element Analysis (FEA) frameworks** to solve complex engineering problems.
- Developed the **FE-LSTM approach**, achieving up to **99% reduction in computational time** for nonlinear simulations of heterogeneous materials.
- Enabled **computationally expensive numerical simulations** to run on standard desktop computers using **FE-LSTM**, significantly reducing reliance on **High-Performance Computing (HPC) resources**.

Technical Skills

FEA / CAE: Altair HyperMesh, HyperWorks, MSC Nastran, Abaqus, ANSYS, Gmsh, Simcoon, Fedoo, Microgen
Structural Analysis: Finite Element Analysis (FEA/FEM), Aerospace Structures, Composite Structures, Structural Analysis, Solid Mechanics, Nonlinear Mechanics, Multiscale Modeling, Heterogeneous Materials, GFEM, Model Condensation, Design Optimization, Stress Analysis
AI / Machine Learning for Engineering: TensorFlow, Keras, scikit-learn, Deep Learning, Hybrid AI–Physics Modeling, Physics-Based Machine Learning, Surrogate Modeling, Simulation Acceleration
Programming / Data: Python, C++, MATLAB, Pandas
Languages: English (Fluent), Arabic (Native), French (Native)

Education

Ph.D. in Mechanical Engineering **2019–2022**
Institut de Mécanique et d'Ingénierie de Bordeaux (I2M) Bordeaux, France
Master of Engineering, Applied Mathematics and Mechanical Engineering **2016–2019**
Bordeaux INP ENSEIRB-MATMECA Bordeaux, France

Awards & Distinctions

CSMA PhD Thesis Award 2022 — Laureate 2023
Computational Structural Mechanics Association (CSMA), Giens, France
Laureate selected among 28 PhD thesis applications. Awarded for PhD research on AI-driven numerical simulation of heterogeneous materials combining artificial intelligence with physics-based modeling. Nominated for the ECCOMAS PhD Thesis Award.